

Which of the following is a nucleotide sequence database?

**Select the correct option**

|                                  |              |
|----------------------------------|--------------|
| <input checked="" type="radio"/> | All of these |
| <input type="radio"/>            | Ensemble     |
| <input type="radio"/>            | NCBI         |

Which of the following is a nucleotide sequence database?

**Select the correct option**

|                                  |              |
|----------------------------------|--------------|
| <input type="radio"/>            | GenBank      |
| <input type="radio"/>            | Ensemble     |
| <input checked="" type="radio"/> | All of these |

Where in the DNA information is stored?

**Select the correct option**

|                                  |                              |
|----------------------------------|------------------------------|
| <input type="radio"/>            | The shape of DNA             |
| <input type="radio"/>            | The sugar-phosphate backbone |
| <input checked="" type="radio"/> | The sequence of bases        |

Howe many components does UniProt Consortium have?

**Select the correct option**

|                                  |   |
|----------------------------------|---|
| <input checked="" type="radio"/> | 4 |
| <input type="radio"/>            | 0 |
| <input type="radio"/>            | 3 |

Which of the following is not a UniProt dataset?

Pathogens

**Select the correct option**

|                       |          |
|-----------------------|----------|
| <input type="radio"/> | Taxonomy |
| <input type="radio"/> | Diseases |
| <input type="radio"/> | TrEMBL   |

The UniProtKB (UniProt Knowledgebase) consists of two types of entries, R

**Select the correct option**

|                                  |                       |
|----------------------------------|-----------------------|
| <input type="radio"/>            | UniRef                |
| <input checked="" type="radio"/> | Un-Reviewed (TrEMBL)  |
| <input type="radio"/>            | Un-Reviewed (Protein) |

What is difference between global and local alignment?

|                                  |                                                                                                    |
|----------------------------------|----------------------------------------------------------------------------------------------------|
|                                  |                                                                                                    |
| <input type="radio"/>            | Global alignments are based on dynamic programming                                                 |
| <input type="radio"/>            | Global alignments are faster to compute than local alignments                                      |
| <input checked="" type="radio"/> | Global alignments aim to align every residue in all sequences with the highest level of similarity |



What is the main purpose of aligning DNA or Protein sequences? (Ch

**Select the correct option**

|                                  |                                                             |
|----------------------------------|-------------------------------------------------------------|
| <input checked="" type="radio"/> | To identify similarities and differences between them       |
| <input type="radio"/>            | To provide information about the structure of the sequences |
| <input type="radio"/>            | To determine the genetic makeup of an organism              |

What is Needleman-Wunsch algorithm used for?

**Select the correct option**



For local alignment



For global alignment

Which of the following cannot be related to multiple sequence alignment?

**Select the correct option**

|                                  |                                                                                                            |
|----------------------------------|------------------------------------------------------------------------------------------------------------|
| <input checked="" type="radio"/> | This method does not contribute much to degenerate polymer                                                 |
| <input type="radio"/>            | Multiple sequence alignment also has applications in designing primers based on multiple related sequences |

All are multiple sequence alignment tools, except

**Select the correct option**

|                       |          |
|-----------------------|----------|
| <input type="radio"/> | T-Coffee |
| <input type="radio"/> | BLAST    |
| <input type="radio"/> | Kalign   |

There is a unique advantage of Multiple Sequence Alignment (MSA) to many pair-wise alignments can?

**Select the correct option**

|                                  |       |
|----------------------------------|-------|
| <input checked="" type="radio"/> | True  |
| <input type="radio"/>            | False |

What type of alignment do you use when comparing two sequences from

**Select the correct option**

|                                  |                              |
|----------------------------------|------------------------------|
| <input type="radio"/>            | Global Alignment             |
| <input checked="" type="radio"/> | Pair wise Sequence Alignment |
| <input type="radio"/>            | Local Alignment              |

Which one of the following isn't a BLAST variant?

**Select the correct option**

|                                  |          |
|----------------------------------|----------|
| <input type="radio"/>            | BLASTN   |
| <input type="radio"/>            | BLASTP   |
| <input checked="" type="radio"/> | TBLASTNX |

In MEGA 11, you can create alignment after ----- the loaded sequer

**Select the correct option**

|                                  |                  |
|----------------------------------|------------------|
| <input type="radio"/>            | Preprocessing    |
| <input checked="" type="radio"/> | Selecting        |
| <input type="radio"/>            | Deleting some of |

MEGA stands for

**Select the correct option**

|                                  |                                          |
|----------------------------------|------------------------------------------|
| <input checked="" type="radio"/> | Molecular Evolutionary Genetics Analysis |
| <input type="radio"/>            | Molecular Evolutionary Genes Analysis    |
| <input type="radio"/>            | Molecular Evolutionary Genes Agreement   |

..... are the symbols for recognizing an alignment.

**Select the correct option**

|                       |             |
|-----------------------|-------------|
| <input type="radio"/> | Amino acids |
| <input type="radio"/> | Residues    |
| <input type="radio"/> | Nucleotides |

Taxa in a same clade are also called .....

**Select the correct option**

|                       |                |
|-----------------------|----------------|
| <input type="radio"/> | Species        |
| <input type="radio"/> | Matching pairs |
| <input type="radio"/> | Gaps           |

MEGA 11 allows to change ----- of phylogenetic tree.

**Select the correct option**

|                                  |           |
|----------------------------------|-----------|
| <input type="radio"/>            | Width     |
| <input type="radio"/>            | Height    |
| <input checked="" type="radio"/> | All these |

PDB allows to sort result by .....

**Select the correct option**

|                       |              |
|-----------------------|--------------|
| <input type="radio"/> | Entry ID     |
| <input type="radio"/> | Release date |
| <input type="radio"/> | Score        |

PDB is a repository of .....

*(Choose the most suitable one)*

**Select the correct option**

|                                  |                       |
|----------------------------------|-----------------------|
| <input type="radio"/>            | Protein reads         |
| <input type="radio"/>            | Protein domains       |
| <input checked="" type="radio"/> | Protein 3D structures |

PDB allows users to download in ..... formats.

**Select the correct option**

|                                  |                                  |
|----------------------------------|----------------------------------|
| <input type="radio"/>            | Protein sequence in fasta format |
| <input checked="" type="radio"/> | All of these                     |
| <input type="radio"/>            | Ligands                          |

More than 40 instability index shows that a protein is

**Select the correct option**

|                                  |             |
|----------------------------------|-------------|
| <input type="radio"/>            | Hydrophilic |
| <input type="radio"/>            | Stable      |
| <input checked="" type="radio"/> | Unstable    |

**GRAVY** stand for

**Select the correct option**

|                                  |                                 |
|----------------------------------|---------------------------------|
| <input type="radio"/>            | Grand average of hydrophobicity |
| <input type="radio"/>            | Grand average of hydrophilicity |
| <input checked="" type="radio"/> | Grand average of hydropathy     |

Which of the following is untrue regarding PSIPRED?

**Select the correct option**

|                                  |                                                                  |
|----------------------------------|------------------------------------------------------------------|
| <input type="radio"/>            | The multiple sequence alignment is derived from a PSI-BLAST dat  |
| <input type="radio"/>            | It uses a combination of evolutionary information and neural net |
| <input checked="" type="radio"/> | It uses a combination of evolutionary information only           |

PSIPRED has been shown to be capable of achieving an average Q3 s

**Select the correct option**

|                                  |       |
|----------------------------------|-------|
| <input checked="" type="radio"/> | 76.5% |
| <input type="radio"/>            | 78.5% |
| <input type="radio"/>            | 88.5% |

For a good homology modeling the sequence identity should be always

**Select the correct option**

|                       |     |
|-----------------------|-----|
| <input type="radio"/> | 25% |
| <input type="radio"/> | 15% |
| <input type="radio"/> | 45% |

35%

The SWISS-MODEL Template Library is searched in parallel both with \_\_\_\_ target-template alignments

**Select the correct option**

|                                  |                   |
|----------------------------------|-------------------|
| <input type="radio"/>            | BLAST and BLOSUM  |
| <input checked="" type="radio"/> | BLAST and HHblits |
| <input type="radio"/>            | BLAST and Uniprot |

When I-TASSER uses a template, it needs to know both the 3D structure or template sequence

**Select the correct option**

|                                  |       |
|----------------------------------|-------|
| <input checked="" type="radio"/> | True  |
| <input type="radio"/>            | False |

C-score is typically in the range of \_\_\_\_\_, where a C-score of higher va

**Select the correct option**

|                                  |          |
|----------------------------------|----------|
| <input checked="" type="radio"/> | $[-5,2]$ |
| <input type="radio"/>            | $[-5,1]$ |
| <input type="radio"/>            | $[-5,3]$ |

Chimera can be used to delete the protein chains and non-stranded res

**Select the correct option**

|                                  |       |
|----------------------------------|-------|
| <input checked="" type="radio"/> | True  |
| <input type="radio"/>            | False |

The primary language used in chimera is \_\_\_\_\_

**Select the correct option**

|                                  |        |
|----------------------------------|--------|
| <input checked="" type="radio"/> | Python |
| <input type="radio"/>            | C#     |
| <input type="radio"/>            | Java   |

Which of the following is not a member database of InterPro?

**Select the correct option**

|                                  |       |
|----------------------------------|-------|
| <input type="radio"/>            | Pfam  |
| <input type="radio"/>            | HAMAP |
| <input checked="" type="radio"/> | SCOP  |

\_\_\_\_\_ are distinct functional, structural or sequence units that n

**Select the correct option**

|                                  |              |
|----------------------------------|--------------|
| <input type="radio"/>            | Binding site |
| <input type="radio"/>            | Superfamily  |
| <input checked="" type="radio"/> | Domains      |

## Question # 1 of 4 ( Start time: 08:29:29 PM, 08 February 2023 )

GMQE (Global Model Quality Estimation) is expressed as a number between \_\_\_\_\_

Select the correct option

Relo

|                                  |                |
|----------------------------------|----------------|
| <input checked="" type="radio"/> | 0 and 1        |
| <input type="radio"/>            | 0.5 to 1       |
| <input type="radio"/>            | Less than 0    |
| <input type="radio"/>            | Greater than 1 |

TM-score \_\_\_\_\_ means a randomly selected protein pair with the gapless alignment taken from

Select the correct option

|                       |         |
|-----------------------|---------|
| <input type="radio"/> | $<0.14$ |
| <input type="radio"/> | $<0.18$ |
| <input type="radio"/> | $<0.17$ |
| <input type="radio"/> | $<0.15$ |

Correct

\_\_\_\_\_ is a recently proposed scale for measuring the structural similarity betw

**Select the correct option**

|                                  |          |
|----------------------------------|----------|
| <input type="radio"/>            | Z-score  |
| <input type="radio"/>            | C-score  |
| <input checked="" type="radio"/> | TM-score |
| <input type="radio"/>            | L-score  |

In Target-template alignment, if the user knows the structure of the template to use for modelling, format (\*) together with the target protein sequence.

**Select the correct option**

False



True



What is the main function of UniProt Archive (UniParc)?

**Select the correct option**

|                                  |                                                                                                         |
|----------------------------------|---------------------------------------------------------------------------------------------------------|
| <input checked="" type="radio"/> | To capture all publicly available protein sequence data and contains all the protein sequence databases |
| <input type="radio"/>            | To provide comprehensive protein sequence and functional information                                    |
| <input type="radio"/>            | To maintain a vast collection of bioinformatics database and services                                   |
| <input type="radio"/>            | To provide a reference set of clusters for all UniProtKB sequences                                      |

The human genome consists of ~ ..... base pairs.

**Select the correct option**

|                                  |             |
|----------------------------------|-------------|
| <input type="radio"/>            | 2.3 billion |
| <input type="radio"/>            | 3.2 million |
| <input type="radio"/>            | 2.3 million |
| <input checked="" type="radio"/> | 3.2 billion |

What is the main function of UniProt Knowledgebase (UniProtKB)?

**Select the correct option**

|                                  |                                                                        |
|----------------------------------|------------------------------------------------------------------------|
| <input checked="" type="radio"/> | To provide comprehensive protein sequences and functional information  |
| <input type="radio"/>            | To maintain a vast collection of bioinformatics databases and services |
| <input type="radio"/>            | To store metagenomic and environmental sequences                       |
| <input type="radio"/>            | To provide a reference set of clusters for all UniProtKB sequences     |

What is the main function of UniProt Metagenomic and Environmental Sequences (UniMES)?

**Select the correct option**

|                                  |                                                                            |
|----------------------------------|----------------------------------------------------------------------------|
| <input type="radio"/>            | To maintain a vast collection of bioinformatics databases and services     |
| <input type="radio"/>            | To provide a reference set of clusters for all UniProtKB sequences         |
| <input checked="" type="radio"/> | To store sequences which are recovered directly from environmental samples |
| <input type="radio"/>            | To provide comprehensive protein sequence and functional information       |

What type of sequences can be found in UniMES?

**Select the correct option**

|                                  |                                                       |
|----------------------------------|-------------------------------------------------------|
| <input checked="" type="radio"/> | Metagenomic and environmental sequences               |
| <input type="radio"/>            | Reference set of clusters for all UniProtKB sequences |
| <input type="radio"/>            | Protein sequence and functional information           |
| <input type="radio"/>            | None of these                                         |